



Department of Material Science and Engineering University of Moratuwa

Design of Door Handle

MT2220 - Ferrous Metals and Alloys

Group B

26/10/2023

Team Members

1. 210101A - DE SILVA G.A.I
2. 210276L - KARUNARATHNE W.K.S.A
3. 210311R - KUMARA R.P.D.R
4. 210345A - LIYANAGE S.W
5. 210494D - PREMAKUMARA H.P.S
6. 210533A - RATHNAMALALA T.N.S
7. 210564T - SANDARU H.W.P
8. 210700J - WICKRAMAGE W.S.E.L

TABLE OF CONTENT.

Page No:

1. Introduction	1
2. Research and Analysis	2
• Material selection	
• Heat treatment	
• Mold and die design.	
3. Designing	3
• Door handle	
• Mold	
4. Heat Treatment Plan	9
5. Conclusion	15
6. References	19

1. INTRODUCTION

Designing a door handle, a modest element in the grand scheme of architecture and interior design, is, in fact, a compelling intersection of functionality and artistic expression. Door handles serve as both practical tools for entry and exit and as aesthetic accents that contribute to the overall ambiance of a space. This design project embarks on a captivating journey to delve into the depths of crafting a door handle that transcends its utilitarian purpose. It aims to be an art form, not just a mechanism for opening doors. In this report, we will explore the multifaceted considerations that go into designing a door handle, touching upon the significance of materials, ergonomics, cultural connotations, and stylistic influences, all of which coalesce to create a door handle that seamlessly combines form and function.

The door handle is, without a doubt, an essential interface between human beings and their architectural surroundings. It is the first point of physical contact with a space, setting the initial tone for the visitor's experience. This project will dissect the door handle's role as a facilitator of access, ensuring its ergonomic qualities prioritize user comfort. Moreover, we will delve into the cultural and historical significance of door handles, recognizing how they can symbolize traditions, meanings, and even societal values. The report will also consider the importance of stylistic influences, understanding how the door handle's design can be tailored to complement architectural themes or stand as a unique statement piece. Through this holistic exploration, our aim is to craft a door handle that harmoniously merges both its aesthetic and practical dimensions, leaving a lasting imprint on those who interact with it and elevating the character of the spaces it graces.

The process of designing a door handle, often underestimated, involves a profound understanding of the intricate balance between the tactile, visual, and functional aspects of this unassuming yet crucial architectural detail. Our endeavor is to create a door handle that not only fulfills its utilitarian purpose but also transcends into a piece of functional art that enhances the aesthetic appeal and user experience of any space. This report will take you on a journey through the creative process, shedding light on the complexities and considerations involved in crafting a door handle that is both inviting and efficient, reflecting a fusion of creativity, culture, and practicality.

2. RESEARCH AND ANALYSIS

2.1 Material Selection

1) Stainless Steel Door Handles



Figure 1 : Stainless Steel Door Handle

Stainless steel is a type of steel generally formed of iron, carbon, and other materials. It is incredibly long-lasting and durable. It's a fantastic choice for a door handle since its strength makes it more difficult to break, adding an extra degree of security. It is also cost-effective, as there are numerous steel door handles available. The look of stainless steel door knobs and handles means that it isn't a popular choice for interior doors.

However, not all steel is stainless, which implies that only a tiny percentage of it is resistant to corrosion and rust. When purchasing a steel door handle, specify that you want a stainless one to assure the product's longevity and minimize accidents. Aside from that, this is one of the most practical and long-lasting door handle materials.

- **Advantages**

- 1) Stainless steel door handles are durable.
- 2) They have proven to be long-lasting.
- 3) Stainless steel door handles are cost-efficient.

- **Disadvantages**

- 1) Steel is not always stainless.
- 2) Stainless steel might not be appropriate for modern houses.

Suitable areas: Stainless steel door handles are often used in residential and commercial settings, particularly in modern or contemporary-style buildings. They may also be used in outdoor or industrial environments due to their durability and resistance to corrosion.

2) Wood Door Handles



Figure 2: Wood Door handle

Solid wood has been a primary construction material for many years. It has proven to be sturdy when used as a door handle. Wood is also excellent for creating a design specifically with a neutral style. Several coat finishes can be added to make it more robust and seamless.

Water and humidity may compromise the structural integrity of this material, making it unsuitable for comfort rooms or outdoor areas. When used in bedrooms, kitchen doors, or living rooms, it has the most extended lifespan. Because it can be carved into any form or size and coated in any color, wood is, however, an excellent material for making custom doors.

- **Advantages**

- 1) Wood has a natural color that can complement the aesthetics of the house.
- 2) It is effortless to clean.
- 3) Wood is durable and long-lasting.

- **Disadvantages**

- 1) Wood is not ideal to use in wet areas or humid areas.

3) Aluminum Door Handles



Figure 3: Aluminum Door handle

Aluminum has a silvery-white sheen that makes it a great design choice for handle materials. Designer door handles get made of this material. All aluminum handles, unlike some steel, do not rust, making them suitable for outdoor use because they can resist weather changes. The only disadvantage of this material is its cost. While some can be pretty costly, when you consider the benefits, it is well worth it in the end. The simple, industrial look of aluminum handles makes them a better choice for outdoor doors or commercial spaces.

- **Advantages**

- 1) Aluminum is effortless and convenient to maintain.
- 2) It is durable and long-lasting.
- 3) Aluminium is rust resistant.
- 4) Lightweight: Aluminium is much lighter than other materials, such as steel or brass, which makes it easier to install and operate.

- **Disadvantages**

- 1) The main disadvantage of aluminum is it is a high price.
- 2) Aesthetic appeal: Aluminium has a more utilitarian appearance than other materials and may not be as visually appealing as brass or stainless steel.

Suitable areas: Aluminium door handles may be ideal for use in residential or commercial settings, particularly in modern or contemporary buildings. They may also be used in outdoor or industrial environments due to their lightweight and low-maintenance properties.

4) Brass Door Handles



Figure 4: Brass Door handle

Brass is excellent for custom-made for both aesthetic and mechanical reasons. Because of its aesthetic value and antibacterial properties, It is a popular choice for decorative applications. Brass is also available in a range of colors, from delicate gold and silver to almost red. These doors handles are affordable and fire-resistant, meaning they are safe and can safeguard the door or anything behind it from fire damage. Brass commonly combines with other metals for specific purposes. These door handles have a lovely, one-of-a-kind look that is appealing.

- **Advantages**

- 1) Brass door handles are beautiful and elegant.
- 2) They are solid and last long.
- 3) Brass has antibacterial activity (Brass, as well as other metals like zinc, copper, and silver, are oligodynamic. This means they have ions that can kill living cells, such as fungi, bacteria, and viruses)
- 4) Brass is also durable and is not prone to corrosion like other materials, so it won't need much upkeep to stay beautiful.
- 5) Brass can withstand high temperatures and is not prone to melting or deforming.

- **Disadvantages**

- 1) Brass door handles can be scraped or dented.
- 2) They can corrode in harsh weather.
- 3) The color of brass fades with time.
- 4) Allergy risk: Some people may be allergic to brass, which can cause skin irritation or other allergic reactions.

Suitable areas: Brass door handles are often used in residential settings, particularly in traditional or classic-style homes. They may also be used in commercial settings where a more formal or elegant look is desired.

5) Zinc Alloy Door Handles (Popular)



Figure 5: Zinc Alloy Door Handle

Getting really popular recently, zinc alloy handles are also referred to as zinc alloy pulls. Furniture hardware suppliers' handles are common in use as part of a variety of furniture pieces. These zinc alloy handles come in various forms and finishes to complement any door, kitchen, or piece of furniture. Furniture handles providers will help you select the perfect one for your needs. Chrome, sandblasting, nickel, matte, oxidized black, pewter, old-fashioned copper, brass, and many other finishes are available.

- **Advantages**

- 1) Zinc door handles are durable.
- 2) They are cost-efficient and quite affordable.
- 3) They are rustproof and corrosion resistant.

- **Disadvantages**

- 1) Zinc is less aesthetically pleasing.
- 2) It is slightly less strong than other materials.

Suitable areas: Zinc alloy door handles may be ideal for residential or commercial settings, particularly in modern or contemporary buildings. They may also be used in outdoor or industrial environments due to their low cost and low maintenance properties.

6) MDF Door Handles

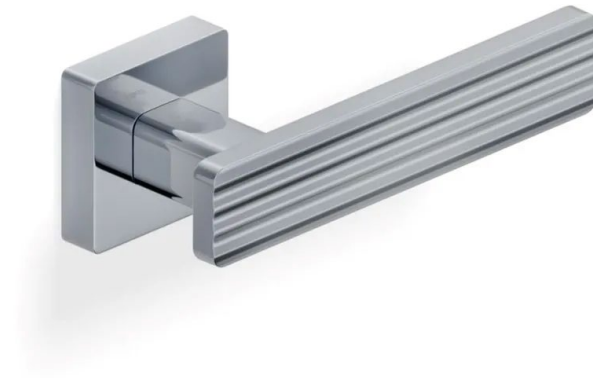


Figure 6: MDF Door handle

Medium Density Fiberboards are a more viable approach to plywood that is gaining popularity due to their durability and low cost. Sand dust is compacted and shaped into a board for its manufacture, subsequently strengthened through several methods. MDF, built with a notched, allows it to function as a door handle. It becomes smoother and more cost-effective this way.

It is not, however, suitable for outdoor use. Because constructed of wood, it is prone to the same problems.

- **Advantages**

- 1) MDF door handles are cost-efficient.
- 2) It has a seamless style.
- 3) These door handles have a smooth surface.

- **Disadvantages**

- 1) It is not ideal to use in humid weather and wet areas.

Selected Material: Stainless Steel

Stainless steel handles are widely used in different industrial cases. There are stainless steel handles of various designs, for example, U shape, bridge shape, foldable and many others.

The quality of the product expected is strong, durable, and versatile handles. These are just some of the reasons stick to stainless steel handles which need quality assurance.

Apart from being durable, it is easy to maintain stainless steel handles. Cleaning these stainless-steel handles won't be a difficult task as compared to other materials. Also, you don't need to add many sophisticated finishes to make stainless steel attractive. Stainless steel handles are quite versatile. They are available in different shapes and designs.

Material Types: 201 Stainless steels, 439 Stainless steel, 304 Stainless steel, 441 Stainless steel

Qualities Expected:

1) Formability

Deformation is important for many handle designs, particularly for appliances. Appliance handles often need to have tight corner radii; in which case you need to prioritize formability. This quality can vary wildly from grade to grade.

2) Cost

Not all stainless-steel grades are created equal in terms of price.

Raw material cost is based in part on availability. Sticking to stainless steel types that are popular and readily available will keep costs down.

3) Corrosion Resistance

The more your product is up against substances that cause corrosion, such as water or solvents, the more corrosion resistance you need. Most popular stainless steels for handle use have high corrosion resistance, but some grades are definitely more corrosion-resistant than others.

4) Appearance

Using the same grade as the appliance or tool is the most obvious solution here, but there are plenty of finishes you can use to adapt the color of the handle. Finishes can transform the look of a handle in practically any way wanted.

5) Durability

Durability is a major concern in certain applications, particularly tools that see rugged use (such as shovel handles). In these cases, you'll want steel that'll hold up for years, not just months.

6) Comfort

Manufacturing needs are only part of the overall equation. You also have to consider what the end user will want from the handle.

A handle that's convenient and easy to grip is a handle that's customer friendly.

Different Qualities Material Types:

1) 304 Stainless Steel (18/8)

This is highly desirable handle steel for multiple reasons. It has great quality and excellent formability. Also, 304 stainless steel has a high nickel content, which gives it a bluish coloring and helps make it more formable.

Of course, that's why it's the most expensive of the steels we're discussing.

There are also spinoffs of 304, such as 304L stainless steel, which has lower carbon content and is easier to weld.

2) 201 Stainless steels

The 201 grade of stainless steel was developed in an attempt to use less nickel than other stainless grades. The lower nickel content makes it cheaper to produce.

It actually has a higher tensile strength than 304 and similar corrosion resistance.

3) 439 Stainless steels

This grade of stainless steel is popular with appliance manufacturers because it is less expensive than both 200 and 300 series steels and often matches the skins of appliances.

4) 441 Stainless Steel

441 stainless steels have great temperature resistance, making it a good choice for oven door handles and similar applications.

Why choose 304 grade stainless steel handles?

304 stainless steel is a ubiquitous choice in modern industry for several compelling reasons. First and foremost, its remarkable **corrosion resistance** makes it a suitable material for applications exposed to **moisture, chemicals, or acidic substances**. This property is invaluable in industries such as food processing, where hygiene is paramount, and in harsh environments like marine settings. Moreover, its robust mechanical properties, including high tensile strength, contribute to its durability and longevity, enabling it to withstand substantial physical stress. Because door handles are always touched by people and will be easy to occur bacteria. If the door handles are made in 304 stainless steel, there is a smaller chance to remain bacteria on the door handles, which is more safe.

In addition to its physical properties, 304 stainless steel offers excellent heat resistance, maintaining its integrity even at elevated temperatures, which is crucial in a wide array of industrial scenarios. Furthermore, its versatility in fabrication, **allowing for welding, machining, and forming, simplifies manufacturing processes**. This adaptability, combined with its polished and stain-resistant surface, lends itself to both **functional and aesthetic applications**, making it a popular choice for architectural and decorative elements in modern industry. Lastly, its wide availability in various forms further enhances its convenience for manufacturers and industrial applications.

The chemical composition of 304 stainless steel (also known as 18-8 stainless steel) is typically as follows:

- Carbon (C): 0.08%
- Chromium (Cr): 18-20%
- Nickel (Ni): 8-10.5%
- Manganese (Mn): 2%
- Silicon (Si): 1%
- Phosphorus (P): 0.045%
- Sulfur (S): 0.03%
- Iron (Fe): Balance

The mechanical properties of 304 stainless steel can vary slightly depending on the specific manufacturing and heat treatment processes, but typical values are as follows:

- Tensile Strength: 515-620 MPa (75-90 ksi)
- Yield Strength: 205 MPa (30 ksi)
- Elongation at Break: 40% (in 50 mm)
- Hardness (Rockwell B): 70-90

2.2 Heat Treatment

What is the heat treatment?

Heat treatment is a group of industrial, thermal, and metalworking processes used to alter the physical, and sometimes chemical, properties of a material. The most common application is metallurgical. Heat treatments are also used in the manufacture of many other materials, such as glass.

Heat treatment involves the use of heating or chilling, normally to extreme temperatures, to achieve the desired result such as hardening or softening of a material. Heat treatment techniques include annealing, case hardening, precipitation strengthening, tempering, carburizing, normalizing, and quenching. Although the term heat treatment applies only to processes where the heating and cooling are done for the specific purpose of altering properties intentionally, heating and cooling often occur incidentally during other manufacturing processes such as hot forming or welding.

Heat treatment is used to improve the properties of materials for a variety of applications, including:

3. Strengthening: Heat treatment can be used to make metals stronger and harder, which is important for many applications such as tools, dies, and machine parts.
4. Improving toughness: Heat treatment can also be used to improve the toughness of metals, which makes them more resistant to impact and fracture. This is important for applications such as aircraft components and automotive parts.
5. Reducing stress: Heat treatment can be used to reduce stress in metals, which can improve their fatigue resistance and dimensional stability. This is important for applications such as springs and precision parts.
6. Improving machinability: Heat treatment can be used to improve the machinability of metals, which makes them easier to cut and form. This is important for many manufacturing processes.

Main stages of heat treatment

1) Heating

To ensure uniform heating of metal during the heating stage, a gradual heating process is essential.

Uneven heating can lead to distortion or cracks in the metal. The heating rate should be chosen based on the metal's heat conductivity, the condition of the metal (whether it has been previously hardened or stressed), and the size and cross-section of the metal. Metals with high heat conductivity heat up faster than those with low conductivity. Hardened or stressed metal should be heated more slowly than unaffected metal, and larger or unevenly shaped parts need slower heating to prevent cracking or warping.

2) Soaking

The soaking stage is a heat treatment process that involves keeping metal at a specific temperature for a period of time to achieve a desired internal structure. The soaking period is determined by the chemical analysis and mass of the metal, as well as the thickness of the largest section for uneven cross-sections.

To perform the soaking stage, the metal should be heated slowly to just below the temperature where the structure will change, and then held until the temperature is consistent throughout the metal. This is known as preheating. After preheating, the metal is more quickly heated to the final soaking temperature. Parts with more complex designs may require layers of preheating to prevent warping.

3) Cooling

The cooling stage of heat treatment is important for determining the desired properties of the metal. Different metals and alloys require different cooling rates and media. Quenching is a rapid cooling process that is usually associated with hardening, but it can also be used to anneal some metals. Not all metals can be quenched safely. The general guidelines are that water can be used to harden carbon steels, oil to harden alloy steels, and water to quench nonferrous metals. However, it is important to consult a heat treatment expert to determine the best cooling rate and medium for your specific metal and application.

Basic heat treatment process

There are 4 main type of heat treatment. These are,

7. Annealing
8. Normalizing
9. Hardening
10. Tempering

Annealing

Annealing is a heat treatment process that alters the physical and sometimes chemical properties of a material to increase its ductility and reduce its hardness, making it more workable. It involves heating a material above its recrystallization temperature, maintaining a suitable temperature for an appropriate amount of time and then cooling. In annealing, atoms

migrate in the crystal lattice and the number of dislocations decreases, leading to a change in ductility and hardness. As the material cools it recrystallizes.

Annealing is often used on metals to make them more formable or machinable. It can also be used to relieve stress in metals, which can improve their fatigue resistance. Annealing is also used on other materials, such as glass and ceramics, to improve their properties.

There are 5 main types of different types of annealing processes. There are,

- Full annealing
- Spheroidizing annealing
- Process annealing
- Stress relief annealing
- Isothermal annealing

1) Full Annealing

Full annealing constitutes a heat treatment technique primarily applied to metals and alloys. It involves carefully heating the material to a specific temperature, maintaining it at that temperature for a defined duration, and subsequently executing a controlled, gradual cooling process. The fundamental purpose of this procedure is to diminish the material's hardness while simultaneously enhancing its ductility. Full annealing is commonly employed following cold working or other procedures that induce hardness and brittleness in the material.

The full annealing process comprises several essential stages. First, the material is heated to a precise temperature, typically within a range of 30-50°C above its upper critical temperature (UCT). The UCT denotes the temperature at which the material undergoes a phase transformation from ferrite to austenite, resulting in the development of a face-centered cubic crystal structure.

The second stage of full annealing involves maintaining the material at the designated annealing temperature for a duration that varies, typically spanning from several hours to several days. The specific duration depends on factors such as the material's thickness and composition. During this period, the material undergoes a vital process known as homogenization, facilitating the even distribution of carbon and other elements throughout its structure. This serves to alleviate any internal stresses or strains present within the material.

The final stage of full annealing encompasses the gradual cooling of the material, usually carried out within a furnace or a controlled environment, ultimately reducing its temperature to that of the surrounding room. This gradual cooling, often referred to as furnace cooling, is instrumental in averting excessive brittleness in the material and significantly mitigating the risk of cracking or warping.

Benefits of Full Annealing:

Full annealing offers several benefits over other heat treatment methods, including:

1. Increased ductility: Full annealing softens the material and increases its ductility, which

makes it easier to form and shape.

2. Improved machinability: Full annealing can improve the machinability of the material, making it easier to cut and shape without damage.
3. Improved machinability: Full annealing can improve the machinability of the material, making it easier to cut and shape without damage.
4. Reduced internal stresses: Full annealing helps to reduce any internal stresses or strains within the material, which can improve its overall strength and durability.
5. Enhanced electrical conductivity: Full annealing can improve the electrical conductivity of some materials, making them more suitable for electrical applications.

2) Spheroidize annealing

Spheroidize annealing is a heat treatment method suitable for steel containing over 0.8% carbon. During this process, components are elevated to temperatures ranging from 1150°F to 1200°F and maintained at this level for a specific duration, which induces a transformation in the microstructure. Essentially, the cementite phase shifts from a lamella (layered) structure to an alpha ferrite matrix interspersed with spheroidal particles of cementite (Fe_3C). Spheroidize annealing is typically employed on components

that have undergone work hardening, thereby enabling further processing, such as rolling for coils or wire drawing. As a result, the resulting product exhibits enhanced ductility and toughness, accompanied by reduced hardness and strength. This annealing procedure is commonly conducted within a protective (endothermic) atmosphere to prevent oxidation and decarburization.

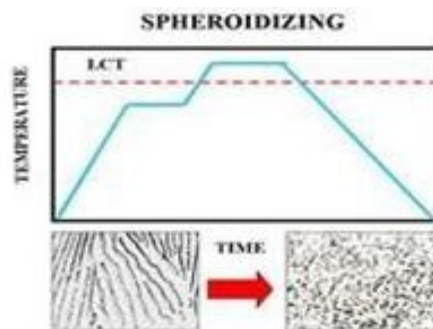


Figure 7: Spheroidizing

3) Process annealing

This heat treatment involves heating steel, or any applicable material, to a temperature below its critical point and maintaining it at that temperature for a sufficient duration before gradually cooling it. The rate of cooling is relatively unimportant since the process occurs at temperatures below the critical range.

The primary objective of this treatment is to decrease the hardness and enhance the ductility of steel that has undergone cold working, thereby facilitating further processing. This method is extensively employed in the treatment of thin sheets and wires.

Components created through cold forming techniques like stamping, extrusion, upsetting, and drawing frequently undergo this treatment as an intermediary step. Moreover, it serves to prevent or minimize scaling and oxidation, particularly when annealed at lower temperatures or in non-oxidizing environments.

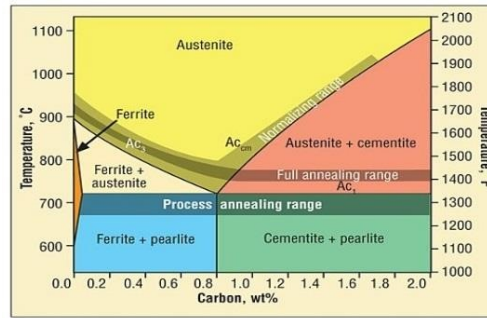


Figure 8: Fe-C Phase Diagram

4) Stress relief annealing

Uneven heating or cooling of steel can lead to the formation of internal stresses, commonly referred to as residual stresses. These residual stresses are frequently generated during welding, where the workpiece is locally heated and subsequently cooled. However, similar stresses can also develop in the workpiece during processes like milling or turning, where high temperatures are produced in the machining zone. Additionally, during the hardening process, quench distortion can result from the residual stresses induced by uneven cooling. Residual stresses, among their various effects, have the potential to reduce the overall strength of the workpiece. Furthermore, if these stresses are suddenly released during a subsequent machining operation, they can cause distortion in the workpiece. Therefore, there are cases where it becomes necessary to alleviate these residual stresses in the workpiece. This is achieved through a method known as stress relief annealing.

In stress relief annealing, the workpiece is subjected to annealing at temperatures below the PSK-line, typically within the range of 550°C to 650°C. The key principle behind stress relief is that the strength of the material decreases as it is heated to higher temperatures. When the yield strength (hot yield

strength) drops below the value of the residual stresses, they can be reduced through plastic deformation. This reduction is facilitated by the movement of dislocations within the material. However, it's important to note that residual stresses can only be reduced to a maximum of the corresponding hot yield point, and they may never be completely eliminated.

After the annealing process, the workpiece must be cooled slowly to prevent the reappearance of stress. In most instances, the workpiece is allowed to cool gradually within the turned-off annealing furnace.

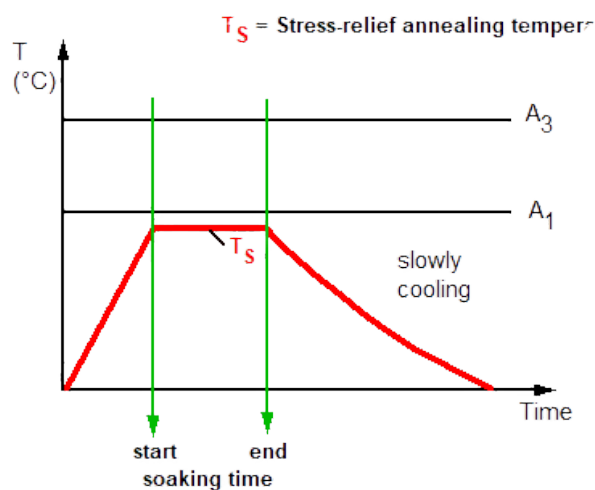


Figure 9: Stress Relief annealing

5) Isothermal annealing

Isothermal annealing is a heat treatment process that effectively alleviates the stresses resulting from both hot and cold plastic shaping, significantly reduces material hardness, and fosters the uniformity of the metallographic structure.

At an industrial scale, complete annealing is typically favored due to its practicality in terms of both time and resources.

The isothermal annealing process encompasses several distinct stages:

- Austenitization
- Critical cooling
- Isothermal transformation of austenite
- Controlled cooling
- Free cooling

The primary goal is to achieve optimal workability of the material and ensure outstanding dimensional stability, even following subsequent heat treatments like carburizing, hardening, tempering, or further hardening.

Steels particularly well-suited for this treatment include case-hardening steels, specialized mold steels, and hypereutectoid steels.

Normalizing

Normalizing in material science is a heat treatment process that is used to refine the grain structure of a metal and improve its mechanical properties, such as ductility, toughness, and strength. It is typically performed on ferrous alloys, such as steel, but can also be used on non-ferrous alloys, such as aluminum and copper.

The normalizing process involves heating the metal to a temperature above its upper critical temperature (A_{c3}), which is the temperature at which the austenite phase begins to form. The metal is then held at this temperature for a period of time, typically 30 to 60 minutes, to allow the austenite to form completely. The metal is then cooled in air to room temperature.

During the cooling process, the austenite transforms back into ferrite and pearlite. The grain size of the ferrite and pearlite is smaller than the grain size of the original structure, which results in a

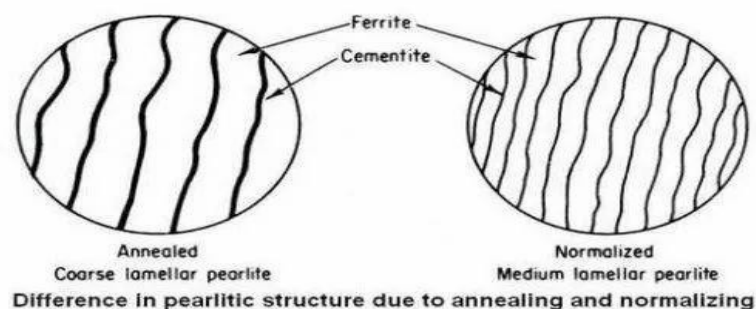


Figure 10: Grain structure comparison

more refined microstructure. This refined microstructure has improved mechanical properties, such as ductility, toughness, and strength.

Normalizing is often used to prepare metals for further processing, such as machining or forging. It can also be used to improve the properties of metals that have been hardened through other processes, such as quenching and tempering.

Hardening

Metal hardening serves as a valuable technique for enhancing a wide range of mechanical properties in metals, with a particular focus on increasing hardness. The objective is to produce a sturdier and more enduring metal product.

During this procedure, alloys undergo heating to temperatures surpassing the critical transformation point of the metal, followed by rapid cooling. This swift cooling serves the purpose of transforming the initially soft material into a structure that is both tougher and stronger. The specific method of cooling, whether through exposure to air or quenching in substances such as oil, water, or other liquids, depends on the composition of the alloy. Subsequently, the hardened materials commonly undergo tempering or stress relief treatments to further enhance their toughness.

In the context of working with steel, employing heat treatments becomes imperative to achieve superior mechanical properties, including increased strength and hardness. The hardening process entails raising the components' temperature above the normalizing threshold and maintaining this temperature for a minimum duration of one hour per inch of thickness. This facilitates cooling at a rate that allows the material to transform into a harder and more robust structure.

Steel, being an alloy of iron and carbon, can also incorporate additional metal elements. Elevating the temperature beyond the critical point leads to the carbon and potentially these other elements entering a solid solution. Subsequent quenching effectively "freezes" this microstructure in place, thereby preparing it for the tempering process to fine-tune its properties.

Tempering

Tempering, also known as drawing, constitutes a heat treatment procedure wherein the materials undergo heating and retention at a specified temperature below the critical point for a designated duration. Subsequently, the materials are cooled to room temperature in a quiescent air environment.

Similar to other heat treatment processes like annealing and normalizing, tempering serves to modify the undesirable mechanical properties of metals, aligning them more closely with the intended application.

Tempering has a broad-reaching impact on the mechanical characteristics of an entire component, spanning from the surface to the core. Nonetheless, in certain cases, partial tempering can be performed using induction equipment.

Tempered metals find utility in applications requiring a specific level of flexibility in their components. Moreover, this heat treatment process is also employed to mitigate the heightened hardness resulting from the welding of components. The localized, elevated temperatures during welding can lead to increased hardness in heat-affected zones. Tempering serves as an effective means to alleviate these areas of excessive hardness. While theoretically applicable to a variety of metals, tempering is most commonly associated with carbon steel, as only a few other metals respond to this heat treatment technique in the same manner as steel.

How does heat treatment enhance the properties of selected material?

After casting the door handle, there are several important properties to focus on. These properties can be enhanced or controlled through various processes after casting, such as heat treatment, polishing, coating, and quality control.

1. **Hardness:** This is the ability of the door handle to resist deformation when a load is applied. It's important for the handle to be hard enough to withstand regular use without deforming. While austenitic stainless steels like 304 are not hardenable by heat treatment, the hardness can be slightly increased through cold working processes such as cold rolling or drawing.
2. **Toughness:** This is the ability of the door handle to absorb energy and plastically deform without fracturing. A tougher door handle is less likely to break or crack when subjected to stress. The toughness of 304 stainless steel is inherently high due to its austenitic structure. Stress Relief Treatment can further improve toughness by relieving internal stresses induced during manufacturing.
3. **Corrosion Resistance:** Door handles are often exposed to various environmental conditions, so they need to be resistant to corrosion. Solution Treatment (Annealing) can enhance the corrosion resistance of 304 stainless steel by allowing for a more uniform distribution of alloying elements.
4. **Surface Finish:** The door handle should have a smooth surface finish for aesthetic reasons and to ensure a comfortable grip. The surface finish is typically improved through mechanical processes such as polishing rather than heat treatment. However, a proper annealing process can provide a clean, scale-free surface that might enhance subsequent finishing operations.
5. **Dimensional Stability:** The door handle should maintain its dimensions and not deform or shrink over time. This is important for the handle to fit properly and function effectively. Stress Relief Treatment can improve dimensional stability by relieving internal stresses that might otherwise lead to distortion.
6. **Wear Resistance:** This is the ability of the door handle to withstand wear and tear from regular use. A wear-resistant door handle will last longer and maintain its appearance over time. Wear resistance in 304 stainless steel is typically improved through surface hardening processes rather than heat treatment. However, a smoother surface finish obtained through proper annealing and finishing processes can provide some improvement in wear resistance.

Next look what are heat treatments available for austenitic stainless steel.

Heat treatment of austenitic stainless steel

- ✓ Austenitic stainless steel, which contains a high concentration of chromium (generally more than 18%) and approximately 8% nickel.
- ✓ It is not hardenable by heat treatment as no phase changes occur on heating or cooling.
- ✓ This is because the presence of Cr, Ni, and other alloying elements in austenitic stainless steel lowers the Ms point (Martensite start temperature) to below room temperature (-30 to -70°C), making the austenitic structure stable.
- ✓ There are several heat treatment processes that can be applied to austenitic stainless steels:
 1. Solution Treatment (Annealing): This common heat treatment for 304 stainless steel improves corrosion resistance and other desirable characteristics, especially when the door handle is exposed to corrosive environments.
 2. Stabilization Heat Treatment: This process is used on certain types of stainless steel that contain titanium or columbium to improve its corrosion resistance.
 3. Stress Relief Treatment: This treatment can remove residual tensile stresses in austenitic stainless steel, improving stress corrosion cracking (SCC) resistance and providing dimensional or shape stability.
 4. Surface Hardening: This process can significantly increase the wear resistance of the door handle, enhancing its durability and longevity.

Coldworking for austenitic stainless steel

Cold working processes such as cold rolling or drawing can increase the hardness of 304 stainless steel. It involves deforming the steel at room temperature, which can make the structure of the steel more compact and increase its hardness.

Note: Processes such as Stabilization Heat Treatment, surface hardening, and cold working may not be necessary for a general-use door handle. Stabilization Heat Treatment is typically used for stainless steels that contain titanium or niobium, which is not usually the case for 304 stainless steel. The high strength of 304 stainless steel may eliminate the need for cold working. Surface hardening can improve wear resistance, but this may not be necessary for a general-use door handle, as the natural wear resistance of 304 stainless steel might be sufficient.

2.3 Mold and Die Design

Some method of metal casting are

- (1) Sand Casting: using sand as the mold material.
- (2) Gravity die Casting: using a permanent metal mold and pouring the metal mold by gravity.
- (3) Pressure Die Casting: using a wax pattern and a ceramic mold.
- (4) Centrifugal Casting: using a rotating mold and centrifugal force.
- (5) Investment Casting : using wax removal method.

I. Sand Casting

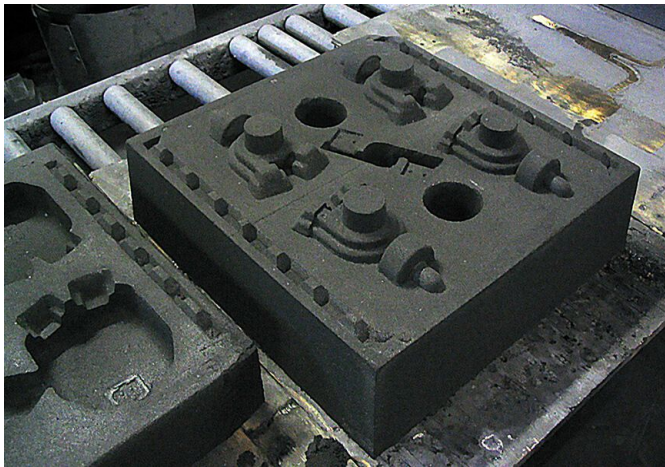


Figure 12: Sand casting 1

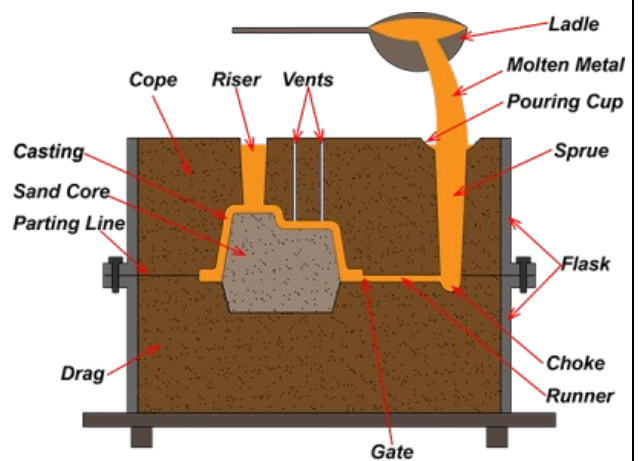


Figure 11: Sand casting 2

The casting process using sand as the molding medium can be term as sand casting or sand molding process. Four stages involve in the process are sand preparation, pattern making, core making (if required), molding and closing.

In normal sand molding procedure, a mold box was prepared to hold sand and the pattern of an exact replica of desired casting. The mold box consists of two parts: the cope (upper part) and the drag (lower part). Then green sand, which is a mixture of sand, clay, and water, is used to create the mold. The sand is loosely packed around the pattern in the drag or cope. After that sand is compacted around the pattern using various tools such as rammers or pneumatic machines (to ensure that mold can withstand the poring of molten metal without collapsing) and then the pattern removed from the mold. Cope and drag brought together and molten metal poured into the cavity created by the mold. After pouring give it time to cool inside the mold. Then the mold is broken open and the casting is removed.

Funchion of molding sand

The basic tool for casting liquid metal to attain required shape is ‘mold’. In this process sand is used as the primary medium for prepare the mold. Ramming it around the pattern and adding a parting compound such as parting powder or oil, is applied to prevent the sand from sticking to the pattern. Once the mold is used it can’t be used for a second time.

Requirement of a good molding sand

a. Refractoriness

The molding sand should have a sufficient refractoriness to withstand the liquid metal temperature while it is pouring. Without sufficient refractoriness, the sand will tend to partially melt and react with the liquid metal forming a very rough sand-fused casting surface, causing rejections.

a. Chemical Resistivity

The molding sand must be inert and not react chemically with metal/alloy being poured into it. While casting more reactive metals like Titanium and Magnesium alloys, special measurement must take to action.

b. Strength with proper binder

The molding sand should develop adequate cohesive among its grain when combined with suitable binder to be able to stay steadily as a mold. It should also adhere to the mold-box or other container when rammed and withstand movement and handling of molds before pouring.

Molding sand should also develop adequate strength with binder to withstand the compressive and erosive force exercised by liquid metal while filing the mold cavity. As the alloy poured has generally high specific gravity they tend to erode, enlarge, and break the mold.

c. Permeability

Liquid metals poured into mold cause reactions with molding ingredients such as binders, additives, and water which reasonable for evolution of gases such as hydrogen, nitrogen, carbon dioxide and steam. These gases tend to trap inside unless the mold has enough porosity to allow these gases and vapors to escape.

d. Surface Finish

The surface smoothness is necessary for appearance and avoiding costly finishing operations. So that this feature is mainly dependent on the condition of the mold which is contributed by the size, shape and distribution of the molding sand grains.

Sand type use in mold for stainless steel products

Zircon Sands,

It contains oxide zircon (Zr_2O) 67% and silica (SiO_2) 33% approximately as a compound. It has the refractoriness among the different base sands. It has very low thermal expansion and high thermal expansion. It can help achieving good surface finish and casting soundness. It is used for high quality casting in alloy steel and precision casting of costly alloys.

II. Gravity Die Casting

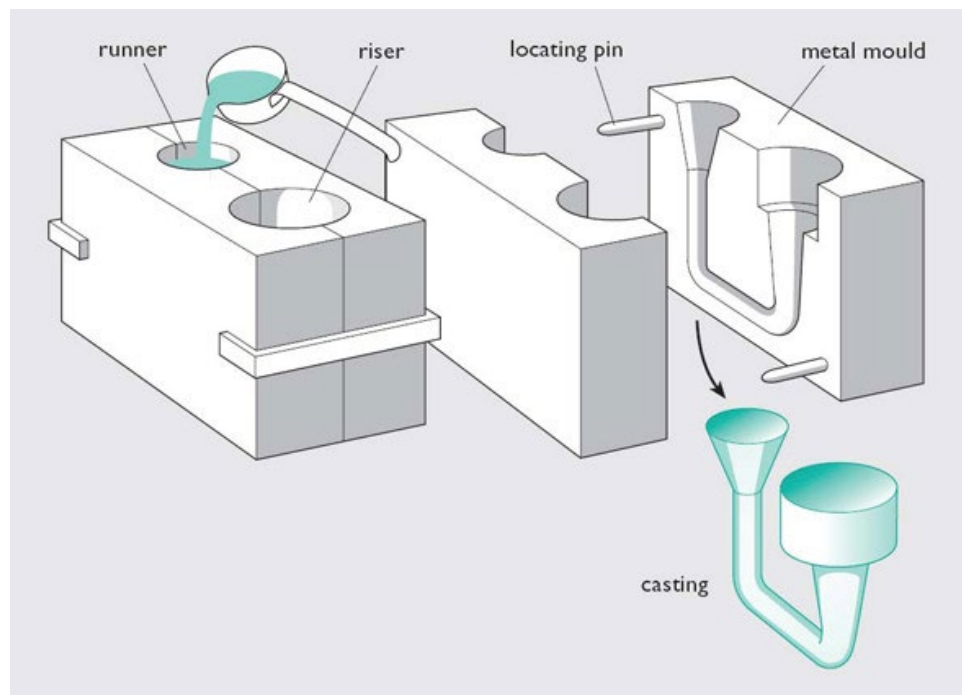


Figure 13: Gravity die casting

Gravity Die casting is a process where molten metal is poured into metallic molds with no application of any external pressure. Sometimes tilting is also required to form a quality product. The molten metal enters the cavity by gravity.

When designing dies for GDC, usage of “cores” is an important issue. Basically, there are multiple materials used to produce cores in various situations such as sand, plaster of paris and metals. This method is suitable for producing high volume of non-ferrous alloy casting of Aluminum, Magnesium, Copper and Zinc base alloys and to a limited extent for cast iron castings. The gravity die casting method can operate manually or by automatic devices. Gray cast iron and steel are the die materials in use. Typical economical volume of production is around 82,500 pieces per die.

The GDC process has many advantages,

- This process is very much suitable for mass production with better reproduction.
- Dimensional accuracy and surface finish are better than sand casting. A minimum wall thickness of 3mm can be cast.
- This process can achieve 20% higher mechanical properties than sand casting due to the faster rate of solidification imparting better grain size.
- The GDC process can be automated and can produce semi-gravity die-castings employing sand or plaster of paris cores for production of interior details.

It has certain disadvantages,

- Limitation of geometry in die and it's difficulty to cast large size high complex shape.
- The casting yield is low when compared to other die casting processes.
- This process is not suitable for steels and super alloys due to their high pouring temperatures.

Requirement for a quality product

i. Die Design

The die must be designed well to dissipate all the heat input to the die before the next casting is poured. It should also have adequate mass to act as a heat sink, to avoid unnecessary heat loss to the extent required during time lapse between castings.

ii. Design of gates and risers

A well-designed gating system must avoid turbulence in metal flow and to reduce incidence of air and inclusions entrapment in the casting. Feeders are required to avoid solidification related defects such as hot tears, shrinkage, micro-porosities etc.

iii. Computer simulations

By doing the casting simulation on the computer including the flow of the molten metal, the heat transfer, solidification and shrinking, and formation of stress can overcome the limitations of uncertainties involved in conventional calculations. Furthermore, by using computer simulation the following can be achieved:

- Optimizing die design
- Optimizing the gating and risering systems
- Prediction of defects such as shrinkage, porosities, hot tears
- Process re-engineering

Materials used for gravity die casting process.

Mostly gray cast iron is often used due to its excellent heat resistance, good thermal conductivity, and durability. Tool steels or alloy steels with high strength and hardness are used for molds that require higher wear resistance and can withstand high-temperature molten metal.

Then a ceramic coating is applied to the mold cavity surfaces to facilitate part removal and to increase the lifespan of the mold. High temperature resistance ceramics and refractory materials, such as silica, zircon, or alumina are used.

Cooling Channels and Venting Methods

cooling channels help control the solidification process and reduce thermal stress while gases can escape during the casting process, preventing defects like porosity. There are simple, straight channels machined into the mold that allow cooling fluid (usually water) to flow through which called Straight-line Channels and serpentine cooling channels follow a winding or serpentine path within the mold which called serpentine channels.

Mold is designed including vented pins and vacuum assist to allow trapped air and gasses to escape,

Vented pins are inserted into the mold to act as venting channels. These pins are typically made of a porous material that allows gases to escape while preventing molten metal from entering the venting channels. In vacuum venting systems use a vacuum to help remove air and gases from the mold cavity. This method can be especially useful for complex or deep drawn parts.

Ejection Mechanism

In many gravity die casting setups, manual ejection is the most common method. It involves using human operators to physically remove the casting from the mold cavity. This method is simple and cost-effective, suitable for low to medium production volumes and relatively straightforward part geometries.

III. Pressure Dye Casting

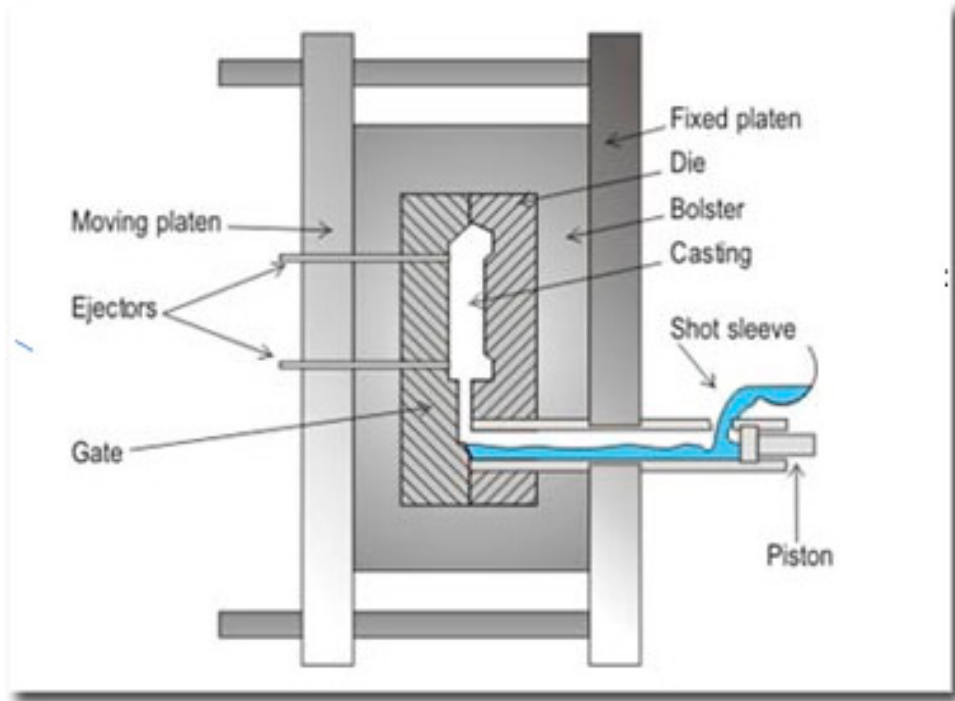


Figure 14: Pressure die casting.

Pressure Die casting is a very quick, reliable, cost-effective and efficient manufacturing method for producing various product forms. The molten metal was forced at high speed, and high pressure into the die cavity. The die consists of moving and stationary half, both of which are mounted to the die casting machine's platens. It has an injection end that uses hydraulics and pressurized gas to move a piston forward, injecting the molten metal into the closed steel die. It also has a clamping end to utilize hydraulics and mechanical toggles to absorb the injection pressure and hold the die shut while the part solidifies. The process can turn molten metal into a solid near-net-shape part in seconds.

The most widely used three metal groups with this die casting process are aluminum, zinc, and magnesium. Pressure die casting can be classified as cold chamber and hot chamber pressure die casting processes.

Advantages of Pressure Die Casting,

- This method can produce large, light-alloy parts in high volumes at great speed and efficiency.
- Low cost compared to other process.
- This process can produce parts with superior surface finish, excellent uniformity, and optimum mechanical properties.
- Create complex and intricate shape in a single piece so it eliminates the need for assembly and welding.

High pressure Die Casting

In here, the molten is injected with high speed and high pressure into the metal mold. The basic machine consists of two vertical platens. Out of them one is fixed and the other is movable. This

mechanism helps the die to open and close. Suitable amount of liquid metal is poured into the shot sleeve and after introducing into the mold cavity. This is basically done using a hydraulically driven piston. After the solidification process complete the die is opened and casting is eventually removed. It can be classified as Hot Chamber process and Cold Chamber process.

Low pressure Die Casting

High quality castings, of aluminum alloys, along with magnesium and other low melting alloys are produced through this method. In this process a metal die is positioned above a sealed furnace containing molten metal. A refractory-lined riser extends from the bottom of the die into the molten metal. Low pressure air is introduced into the furnace. Due to this the liquid metal rise up the tube and enter the die cavity with low turbulence. After the metal has solidified, the air pressure is released. This makes the metal still in liquid state in the riser tube fall back in to the furnace. After subsequent cooling, the die is opened, and the casting extracted.

Material Used

Tool steels are among the most widely used materials for pressure die casting molds due to their excellent combination of hardness, toughness, and wear resistance. Common tool steel grades include H13, P20, and D2. These materials are often heat-treated to achieve the required hardness. Stainless steel grades, such as AISI 420, are used when corrosion resistance is required. They are suitable for casting certain non-ferrous alloys and are known for their durability.

Cooling channels and venting mechanism

Straight line channels and Serpent channels are widely used in both pressures die casting and gravity die casting. There are two primary types of cooling channels: straight-line channels, which are simple and direct channels allowing cooling fluid (usually water) to circulate, and serpentine channels, which wind and snake through the mold to provide more uniform cooling.

To further enhance the casting process, molds are designed to include vented pins and vacuum assist mechanisms. Vented pins are carefully placed within the mold to act as venting channels. They are typically constructed from porous materials, permitting gases to escape while preventing the intrusion of molten metal. Vacuum venting systems, on the other hand, utilize a vacuum to efficiently remove air and gases from the mold cavity. This method is particularly beneficial to produce complex or deep-drawn parts, ensuring that trapped air or gases do not compromise the quality of the final castings.

Ejection mechanism

For castings that are challenging to remove manually due to tight clearances or complex shapes, a tapping or vibrating method may be employed. This involves tapping the mold or using a vibration source to help release the casting by reducing friction between the casting and the mold walls.

Hydraulic or Pneumatic Ejection: In some cases, especially when producing complex or large castings, hydraulic or pneumatic ejection systems can be used. These systems use hydraulic or pneumatic cylinders to push the casting out of the mold. They are more suitable for high-production environments and for castings with intricate geometries.

IV. Centrifugal Casting

The term "centrifugal casting" encompasses a set of casting processes and techniques that leverage the centrifugal forces generated during rotation to uniformly distribute molten metal to the outer periphery of circular molds. These centrifugal forces play a crucial role in maintaining the molten metal at the mold's edge for an adequate duration, allowing it to solidify and ultimately yield a high-quality material capable of withstanding even the most extreme industrial conditions. In industries where the most rigorous conditions prevail, it is imperative that the components or parts being used exhibit the highest possible quality. The presence of any inherent flaws in either the design or the manufacturing process can lead to failures during real-world applications, resulting in a multitude of problems and challenges.

The centrifugal casting process commences by introducing liquid metal into a high temperature die that is set into rotation. The orientation of the die, whether on a vertical or horizontal pivot, depends on the specific design requirements. During this process, centrifugal forces come into play, compelling the liquid metal to disperse within the mold under pressures that can exceed several times the force of gravity. This applied pressure serves to eliminate the formation of cracks and other microscopic and macroscopic defects, akin to the effects observed in conventional die casting, where pressure aids in reducing product defects. As the die fills, the denser liquid metal is driven towards the center of the rotating die. Simultaneously, sound metal solidifies directionally from the outer diameter toward the core, while less dense material and impurities are displaced toward the outer diameter. Upon completion of the solidification process, the finished part is extracted from the die, and any remaining impurities along the part's boundaries are removed through machining operations. The result is a high-quality material devoid of internal voids or air bubbles, making it suitable for a wide range of demanding applications.

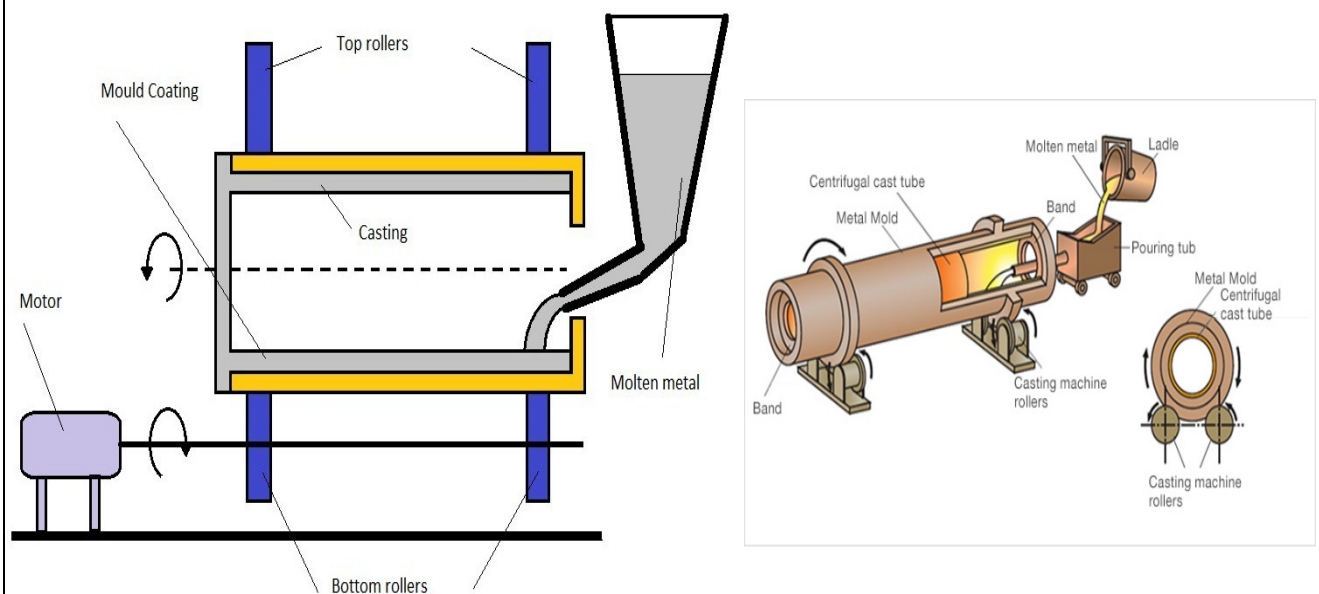


Figure 15: Centrifugal Casting

The centrifugal casting process offers several notable advantages,

Ease of Casting Thin-Wall Cylinders: It provides a practical and resource-efficient method for producing thin-wall cylinders, a task that can be challenging and resource-intensive through other casting methods.

Enhanced Microscopic Qualities: Centrifugal casting results in products with improved microscopic qualities, leading to greater durability and longevity.

Cost-Efficient Complex Castings: This process can efficiently produce complex castings at a lower cost without compromising product quality, making it a cost-effective option for intricate components.

Resistance to Atmospheric Corrosion: The cast products exhibit high density and fewer line defects, contributing to their ability to resist atmospheric corrosion, making them suitable for applications exposed to harsh environmental conditions.

Easy Impurity Removal: The centrifugal casting process simplifies the removal of impurities from the final products, ensuring a high level of purity and quality in the finished components.

It's certain disadvantages,

- Products with small inner diameter have the same impurities.
- Not suitable for products that don't have a circular shape.
- Limitation of the size.

Material Used

Gray cast iron is a popular choice for centrifugal casting molds when producing steel products. It offers excellent heat resistance, good thermal conductivity, and durability. Cast iron molds are well-suited for casting various steel alloys. Certain types of steel, including alloy steels, can be used as mold materials in centrifugal casting. These steel molds are suitable for casting steel products and can provide good wear resistance and dimensional stability. Copper and copper alloys like bronze are sometimes used for specific applications involving steel casting. They offer good thermal conductivity and are commonly used for centrifugal casting of steel products like bimetallic tubes.

Cooling channel and Venting Mechanism

Straight-Line Channels are simple channels or passageways machined into the mold, allowing cooling fluid (usually water) to circulate. Straight-line channels are the most straightforward and common cooling channel design. In some centrifugal casting molds, spiral-shaped cooling channels are used. These channels wrap around the inside of the mold, promoting even cooling and reducing temperature gradients within the mold. Curved or helical channels follow a spiral or helix pattern along the interior of the mold. They help achieve more uniform cooling, especially in molds with complex geometries.

Ejector pins are mechanical pins that can be used to push the casting out of the mold once it has solidified. These pins also serve as venting channels, allowing air and gases to escape during the filling of the mold cavity. In some cases, gas capture systems are used to collect and filter gases emitted during the casting process. These systems help control gas emissions and maintain a cleaner working environment.

Ejection Method

Manual Ejection: Manual ejection is a straightforward and commonly used method in centrifugal casting. In this approach, human operators physically remove the casting from the mold after it has solidified. This method is practical for low to medium production volumes and for parts with straightforward geometries.

Vibration: In some cases, vibration can be employed to assist with the ejection of the casting. Vibrating the mold or the casting can help loosen it, making it easier to remove. This method is often used for castings with complex shapes or tight clearances.

Hydraulic or Pneumatic Ejection: Hydraulic or pneumatic ejection systems may be used in centrifugal casting for parts that are more challenging to remove manually. These systems use hydraulic or pneumatic cylinders to push the casting out of the mold. They are particularly suitable for high-production environments and complex geometry.

V. Investment Casting

Investment casting, also known as lost-wax casting, is a highly precise metal casting process that creates intricate and detailed parts. The process starts with the creation of a wax or plastic pattern, which is then attached to a gating system. This assembly is coated with a ceramic shell through a series of layers. Once the ceramic shell is hardened, it is heated to remove the wax pattern, leaving a hollow mold. Molten metal is poured into the mold, filling the cavity left by the wax. After solidification, the mold is cooled, and the ceramic shell is removed. The final castings are separated from the gating system and undergo finishing processes for shape and surface quality. Investment casting is favored for its ability to produce high-precision parts with fine details and is used in various industries, accommodating a range of metals for different applications.

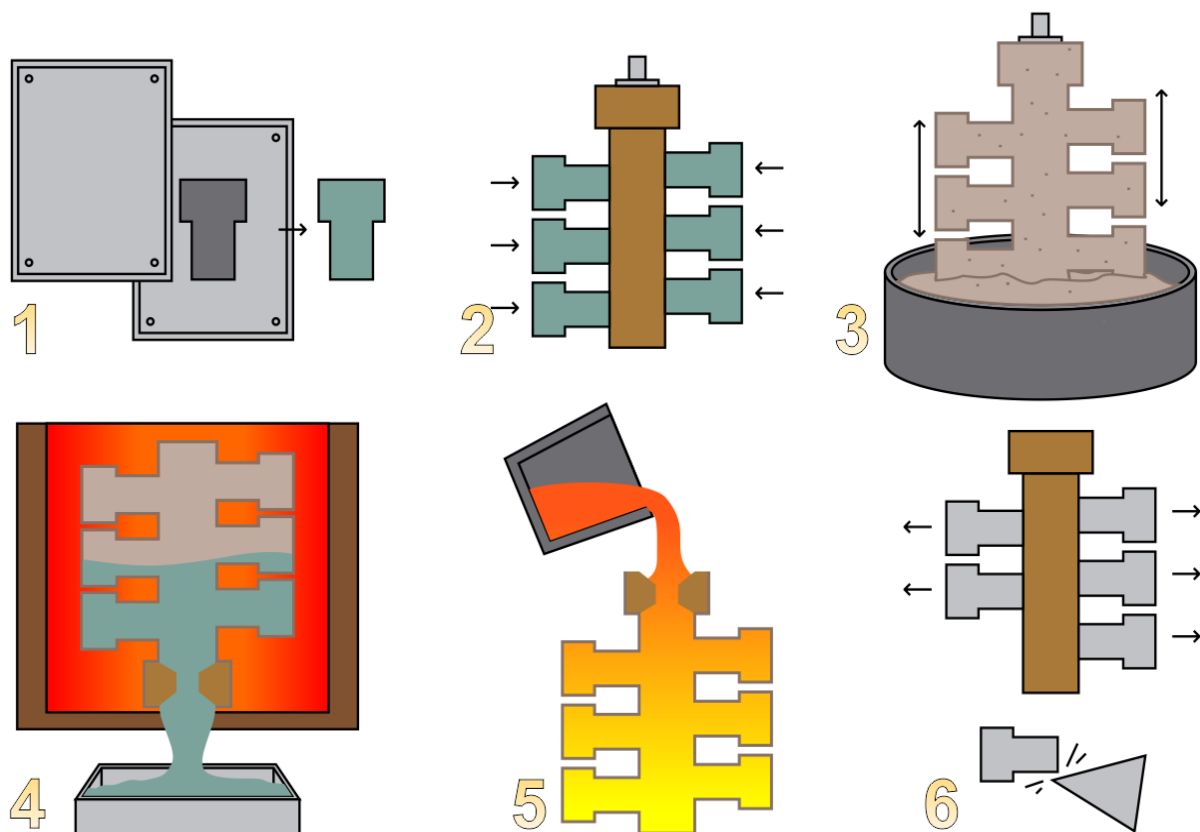


Figure 16: Investment Casting procedure

Procedure

a. Creating the Pattern

The process begins with the design of the part to be cast. This design includes all the intricate details, dimensions, and specifications necessary for the final component. Most of the time this work is conducted through computer-based applications. Then the designed object is made from various materials, with wax (typically microcrystalline or paraffin wax) which is the most common choice due to its ease of use and ability to capture fine details. The pattern can be created by methods such as injection molding, machining, or hand-carving. Sometimes 3-D printing is also used.

b. Mounting the Wax Pattern and Creating the tree

The individual wax patterns are then attached to a gating system, which typically includes a sprue (a main channel), runners (additional channels), and gates. The gating system helps control the flow of molten metal into the mold.

c. Creating the mold Shell

Each wax assembly is carefully inspected to ensure it accurately represents the desired part and that there are no defects or imperfections that could affect the final casting. Multiple wax assemblies can be grouped together on a common sprue to create clusters of patterns. This step streamlines the casting process and maximizes the use of the mold. The wax patterns or clusters are then coated with a ceramic shell through a series of dipping and drying stages (slurry). The ceramic shell will eventually become the mold for the casting.

d. Wax Removal

The assembled tree with the ceramic shell is subjected to the de-waxing process to remove the wax from the mold. This can be done through steam de-waxing (using an autoclave and steam) or flash fire de-waxing (using a high-temperature furnace). The removal of the wax leaves behind a cavity in the ceramic shell, which will be filled with molten metal.

e. Melt and Cast

After de-waxing, the ceramic shell is preheated in the furnace to remove any remaining traces of wax and moisture. This step ensures that the mold is ready to receive the molten metal. Then the preheated ceramic shell is filled with molten metal through a pouring process. The metal is allowed to solidify within the ceramic mold.

f. Final Operations

After solidification, the entire assembly is allowed to cool down. The ceramic shell becomes brittle during this step. Once the casting has cooled, the ceramic shell is broken away, revealing the metal casting inside. This is often done through mechanical means like vibrating or sandblasting. Then the metal casting may require additional finishing processes such as grinding, polishing, machining, or heat treatment to achieve the desired final shape and surface finish.

Advantages of Investment Casting

- **Complex Geometries:** Investment casting is capable of producing intricate and highly detailed parts with complex shapes that would be difficult or impossible to achieve through other manufacturing methods.
- **Tight Tolerances:** It allows for tight dimensional tolerances and high precision, making it suitable for parts where accuracy is critical.
- **Excellent Surface Finish:** Investment casting produces parts with a smooth surface finish, reducing the need for additional finishing processes.
- **Wide Range of Materials:** It can be used to cast a wide range of materials, including various metals and alloys, such as stainless steel, carbon steel, aluminum, and even superalloys for high-temperature applications.
- **Reduced Material Waste:** This method minimizes material waste because the excess material is primarily the ceramic shell, which can be reused to a certain extent.
- **Design Flexibility:** Design changes can be implemented relatively easily, especially when using 3D-printed wax patterns, allowing for rapid prototyping and iteration.

Disadvantages of Investment Casting

- **High Costs:** Investment casting can be expensive due to the multiple steps involved in the process, including pattern creation, ceramic shell formation, and the need for skilled labor.
- **Time-Consuming:** The process is time-consuming, as it involves multiple stages of coating, drying, and curing the ceramic shell, which can lead to longer lead times.
- **Limited Size and Weight:** Investment casting is less suitable for large and heavy parts due to the limitations of the investment casting facilities and the challenges of handling larger ceramic shells.
- **Material Limitations:** While investment casting can work with various materials, certain alloys may have limitations due to their specific melting and solidification characteristics.
- **Fragile Ceramic Shell:** The ceramic shell is fragile and can be easily damaged during handling or transportation, requiring care in these stages of the process.

Cooling Channels and venting mechanism

Cooling channels are an important part of the investment casting process, and they are used to regulate the cooling of the molten metal inside the mold. Proper cooling is essential to achieve uniform solidification and prevent defects in the casting. Cooling channels are integrated into the mold design during the pattern creation stage. Channels are often incorporated into the gating system or core of the pattern.

These channels are often designed to provide passive cooling, allowing heat to dissipate naturally. The shape and placement of these channels should be well-thought-out to ensure even cooling throughout the mold. The choice of material for the cooling channels is crucial. They need to be made from materials that can withstand the high temperatures involved and should be thermally conductive to facilitate efficient heat dissipation.

Vent channels are incorporated into the mold design and connect to the exterior of the mold. These channels allow air and gases to escape as the mold fills with molten metal. Vent channels are strategically positioned throughout the mold to ensure that air can escape from all areas as the metal flows in. They are often located near the top or at the ends of the mold to facilitate the escape of gases. The size of the vent channels is critical. They must be large enough to allow air and gases to escape but not so large that they compromise the integrity of the mold.

Ejection Method

Vibration:

Vibratory methods involve the use of a vibrating table or machine to gently vibrate the ceramic shell, causing it to loosen and separate from the casting. This can be effective for removing the shell without causing damage to the casting.

Ultrasonic Ejection:

Ultrasonic ejection utilizes high-frequency vibrations to help separate the ceramic shell from the casting. This method is particularly useful for intricate and delicate castings.

Thermal Ejection:

In some cases, applying controlled heat to the ceramic shell can help soften the binder material, making it easier to release the casting. This is especially useful when the ceramic shell contains wax-based binders. **Sandblasting:** A controlled sandblasting process can be used to remove the ceramic shell from the casting. This method is suitable for larger and sturdier castings.

3 DESIGNING

3.1 Door Handle

1. Material selection

As mentioned in the previous sections we chose **stainless steel (304)** as the most suitable material for a door handle design. We hope to justify the answer by giving the following points.

- **Corrosion resistance**

It shouldn't come as a surprise to learn that stainless steel handles are resistant to corrosion. Stainless steel is an iron alloy that contains chromium (as well as nickel in some types of stainless steel). The presence of this chromium protects it from corrosion. You can use a stainless-steel handle in a humid environment without fear of it rusting or corroding.

- **Aesthetics**

Most people will agree that stainless steel handles look nice. They offer a consistent bright and shiny appearance. Other types of handles often have a dull or matte appearance, resulting in a lower level of aesthetics. Stainless steel handles offer a higher level of aesthetics thanks to their bright and shiny appearance that's consistent throughout.

- **Hygienic surface**

One of the reasons stainless steel door handles are hygienic is because the material generally doesn't have pores. Bacteria with dirt rarely survive, while door handles are easily disinfected. Due to this benefit, stainless steel door handles are usually used on doors in sanitary and hygiene conditions. Even surgical tools and medical equipment are usually made of it, because of its ability to be sterilized and durable. Properly mirror polished stainless steel door handles are even more hygienic and can be successfully persistent in tough environments.



Figure 17: Different material use for door handles.

2. Ergonomic design

Ergonomic considerations are essential to ensure the door handle is comfortable and easy to use for a wide range of users. A well-designed door handle should cater to the comfort and convenience of users. Key aspects of ergonomic design include,

- **Grip Shape**
The handle should have a comfortable grip shape, accommodating various hand sizes and shapes.
- **Lever vs. Knob**
Choose between a lever handle or a knob, considering user preferences and accessibility requirements.
- **Height and Placement**
Position the handle at an appropriate height, adhering to accessibility guidelines.



Figure 18: Types of door handles

So, in our design we hope to make the door handle completely with stainless steel in a **pull handle style**. Since we should decide the height and placement based on the application, we hope to consider an **interior residential door handle** in which a **normal wooden door** is placed.

3. Aesthetics

Aesthetics play a pivotal role in the design of a door handle, and the selection of stainless steel contributes significantly to the visual appeal and overall cohesiveness of the space.

- **Finish**
The finish of the door handle is a critical aesthetic element that significantly influences its visual appeal. Various finish options are available, each with its unique character. For example, a polished finish offers a high-gloss, mirror-like surface that adds an aura of luxury and sophistication, making it suitable for modern and contemporary designs. In contrast, a brushed

finish provides a textured, linear appearance that softens the reflective quality of stainless steel, creating a more subdued, matte look. A satin finish strikes a balance, offering a smooth, semi-reflective surface that complements a wide range of interior styles. Antique or distressed finishes create a vintage or rustic appearance, adding character and charm to the door handle, making them ideal for traditional or eclectic designs.

- **Design Details**

The design of the handle itself is another crucial element to consider in the aesthetics of a door handle. The shape of the handle can range from simple and sleek to ornate and decorative, depending on the overall design goals. Texture can be incorporated, such as grooves, patterns, or a combination of smooth and textured surfaces, adding depth and interest to the handle. Size and proportion are essential considerations, ensuring that the handle's dimensions are proportionate to the door and the space; oversized handles can make a bold statement, while smaller handles can offer a more delicate look. Aesthetic consistency is vital, as the handle's design should align with the overarching design language of the door, the room, and the entire building, creating a harmonious environment.

- **Color**

While stainless steel handles are typically silver or gray, color variations can be explored to create specific looks. PVD coatings, such as gold, rose gold, black, or bronze, can be applied to stainless steel handles to introduce a range of color possibilities. The choice of color can create a striking contrast with the door's finish or the surrounding environment, drawing attention and serving as a focal point in the design.



Figure 19: Door handle designs

Considering all the aesthetic factors we hope to stick to the **silvery finish** in stainless steel **without applying any color pigments** which will give a majestic look with a polished wooden door.

4. Strength

When considering the strength of a door handle design, several important factors should be considered to ensure the handle can withstand various loads and usage conditions. Here are the key considerations,

- **Material thickness**

The material thickness is a critical aspect of ensuring the strength of a stainless-steel door handle. It's important to determine the appropriate thickness by considering the expected mechanical stress and loads. Thicker material is generally stronger and less prone to bending or deformation. Engineers may perform calculations and simulations to assess the required thickness based on factors such as the handle's dimensions and intended use. While thicker material enhances strength, it's essential to strike a balance between strength and aesthetics. Excessively thick handles can appear bulky and unattractive, so designers often aim for a thickness that combines both the desired aesthetic look and the necessary structural integrity.

- **Reinforcement**

Depending on the handle's design, length, and expected loads, internal reinforcements may be necessary. These reinforcements can take the form of strengthening ribs, support bars, or inserts strategically placed within the handle. Their purpose is to enhance the door handle's structural integrity, making it less susceptible to deformation under stress. Reinforcements should be seamlessly integrated into the handle's design. They should not compromise the overall aesthetic appeal of the handle. Careful consideration of their placement and design ensures that the handle retains its visual attractiveness while gaining added strength.

- **Load capacity**

Load capacity is the handle's ability to withstand the mechanical loads exerted during normal use. This includes pulling, pushing, and twisting forces. To determine the load capacity, it's essential to consider the type of door and its usage. Commercial entrance doors, for instance, will experience significantly higher loads than residential interior doors. The handle design should be tailored to the specific application. To ensure that the handle meets the desired load capacity, manufacturers may conduct load testing and obtain relevant certifications. These tests simulate various types of mechanical loads to verify that the handle can withstand them safely. Compliance with standards, such as those set by ANSI (American National Standards Institute) or other industry-specific organizations, may be necessary to demonstrate load capacity.

So, for the design, we'll consider a **standard frond wooden door** for the load capacity. Due to the **superior procedures followed in heat treatments**, no additional reinforcements are required. After few observations, we were able to figure out the best thickness or the diameter for the handle as **200 mm** (overall)

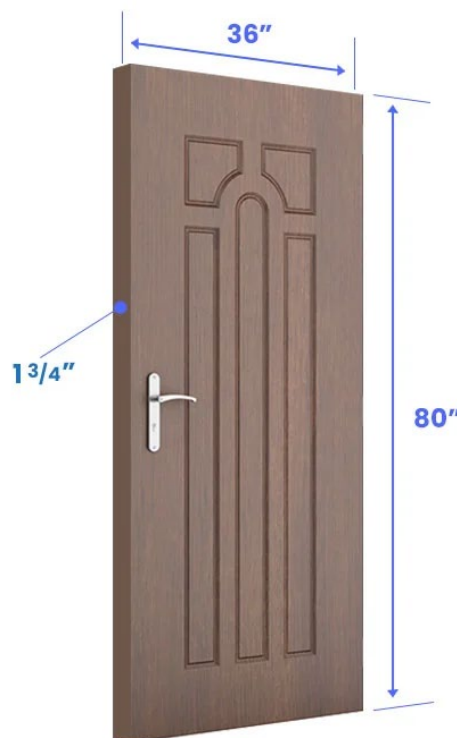


Figure 20: Dimensions of a standard front door

5. Ease of use

When designing a door handle with user-friendliness in mind, it's crucial to consider several factors that enhance the handle's ease of use and overall user experience. Here are three main points that should be considered,

- **Ergonomic Design**

A pivotal aspect of designing a user-friendly door handle is prioritizing ergonomic design. This entails creating a handle that offers a comfortable grip, accommodating various hand sizes and shapes. Ergonomics also extends to deciding between a lever handle or a knob, with lever handles often being the more accessible choice for a wider range of users. Proper placement of the handle at an appropriate height is vital, adhering to accessibility guidelines such as those set by the Americans with Disabilities Act (ADA). Additionally, including a non-slip texture or grip on the handle's surface enhances usability, particularly in environments where users may have wet or gloved hands. An ergonomic design ensures that the door handle is easy to grasp and manipulate, promoting a comfortable and accessible user experience.

- **Smooth and Reliable Operation**

A user-friendly door handle should provide a smooth and reliable operation that allows users to open and close the door effortlessly. This involves ensuring that the internal mechanism of the handle is of high quality and built to last. A well-designed mechanism prevents issues such as jamming, sticking, or squeaking, which can lead to user frustration. Minimizing friction within the handle reduces the effort required for operation, and regular lubrication and maintenance can help maintain its smooth functionality over time. Quiet operation is also essential, as a door handle should not produce excessive noise when in use, particularly in residential settings, where occupant comfort and tranquility are priorities.

- **Accessibility and Inclusivity**

Designing a user-friendly door handle extends to creating a product that is accessible and inclusive to all users, regardless of their physical abilities. Adhering to accessibility standards and regulations, such as those outlined by the Americans with Disabilities Act (ADA), is crucial. This may entail specific height requirements, the use of lever handles, and the provision of appropriate clearances around the handle. User-friendliness also involves ensuring that the handle is designed to require minimal effort for operation, which is particularly important for individuals with limited strength or mobility. Additionally, providing clear signage and instructions can guide users on the handle's operation, especially in public spaces where diverse user needs should be accommodated. An inclusive and accessible design ensures that the door handle is user-friendly for everyone.

So, we thought of **not including any mechanical parts** for the door handle which will lead to wearing issues with usage. So, a **static solid door handle** is concluded.

Considering all the above factors we can present the final design as follows.

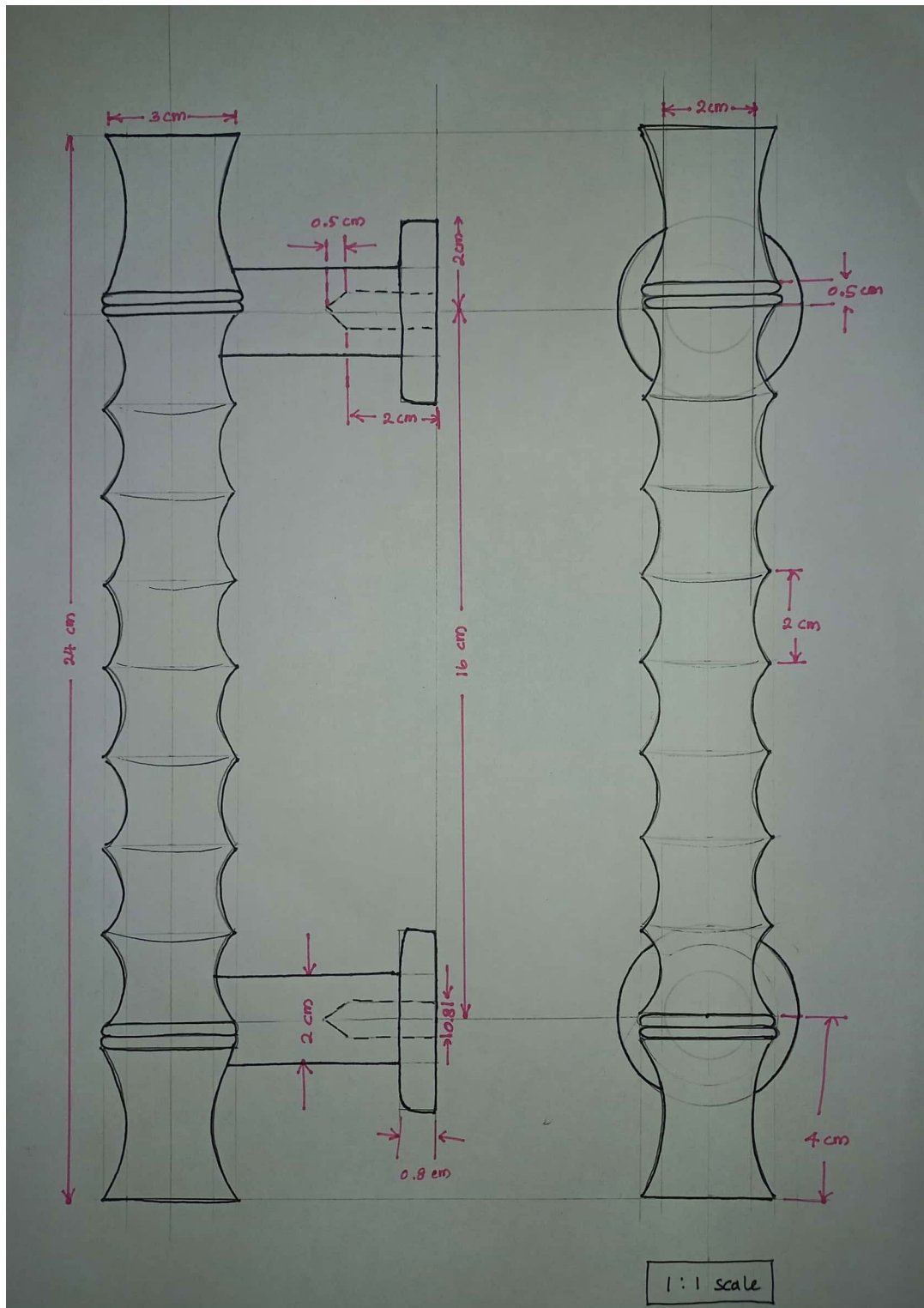


Figure 21: Hand drawing of the design

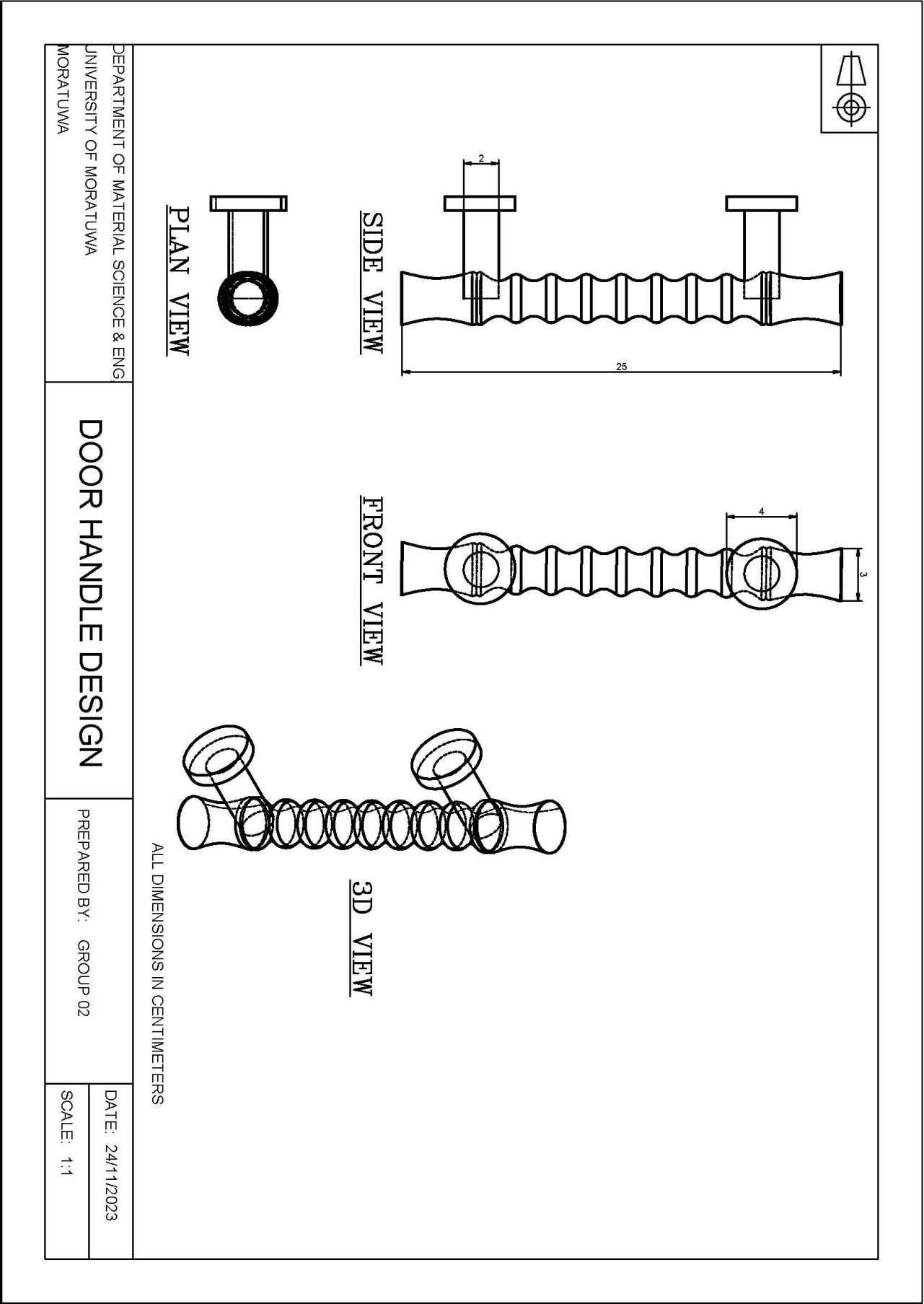


Figure 22: CAD drawing of the design

3.2 Mold / Die Design

There are many methods to manufacture the die/mold for casting and shaping processes considering the properties of materials and the shape of the product. For our application, Stainless Steel was chosen as the material to design the door handle.

The choice of the best casting method for stainless steel door handles depends on various factors, including the complexity of the handle's design, the required precision, the desired surface finish, and the production volume.

Due to the high melting point and other properties of stainless steel, we decided to use investment (lost wax) casting method. The following advantages can be gained by using investment casting for our production.

- High precision, making it suitable for intricate and detailed handle designs.
- Excellent surface finish, reducing the need for extensive post-casting finish.
- Low material waste.

General method and the idea about investment casting is discussed under the Mold/Die research analysis section.

Procedure of casting for door handle is as follows.

- **Pattern Creation -:** Detailed and dimensionally accurate replica of the door handle is created by 3D printing. Since the production is done in lower scale, creating wax pattern by 3D printing is more accurate and efficient. Firstly, the pattern is created by the software SOLIDWORKS. Then the pattern was printed using wax in SL 3D printer.
- **Assembly -:** This step includes attaching wax patterns(model) to wax sprue that will serve as pathways for molten metal to flow. Here 4 patterns of door handles are assembled to create a pattern tree. After attached it is called 'Wax tree' (Pattern Tree).
- **Investment (Mold) Creation-:** Create a ceramic shell or investment mold around the wax model. This involves dipping the wax tree into a ceramic slurry (binder and ceramic material) multiple times, allowing each layer to dry and harden.

After dipping into the ceramic slurry, it should be dipped into stucco several times(6-8 times). Stucco is made with Portland cement, Lime(CaO), Sand, Water, and additives to enhance specific properties. The shell should be thick enough to withstand the pouring of molten metal. (Melting range of 304 stainless steel). The shell now runs through our drying system for 24 to 36 hours before removing the wax from the shell.

- **Dewaxing (Wax Removal)-:** The wax inside the ceramic mold needs to be removed. This is done by heating the entire assembly, causing the wax to melt and drain out, leaving behind a cavity in the mold that matches the shape of the door handle. The temperature for this step can vary but is typically in the range of 160-200°F (71-93°C) to melt and remove the wax.

- Preheating-: The ceramic mold, now empty of wax, is preheated to a high temperature to ensure it can withstand the high temperatures of the molten metal without cracking. The preheating temperature for the ceramic mold is crucial, typically in the range of 1,800-2,200°F (982-1,204°C) (for stainless steel), depending on the type of metal being cast. The actual melting and pouring of the metal occur at its specific melting point, which varies by metal type (e.g., stainless steel around 2,500-2,600°F or 1,371-1,427°C).
- Casting-: Molten stainless steel is poured into the preheated ceramic mold. The metal fills the cavity left by the wax pattern.
- Cooling -: Allow the metal to cool and solidify within the mold. The cooling process may take some time, depending on the thickness of the part.
- Shell removal -: Once the metal has completely solidified and cooled to the touch, the ceramic shell is removed. This is done through vibrating using a vibrating table.
- Cutting and Finishing-:The door handle cast from stainless steel is attached to a sprue, which is removed by cutting or sawing. Any rough edges or imperfections are carefully removed through cutting, grinding, and other finishing processes.
- Transfer to heat treatment processes -: For further enhancement of properties casted door handle is directed to heat treatment processes.

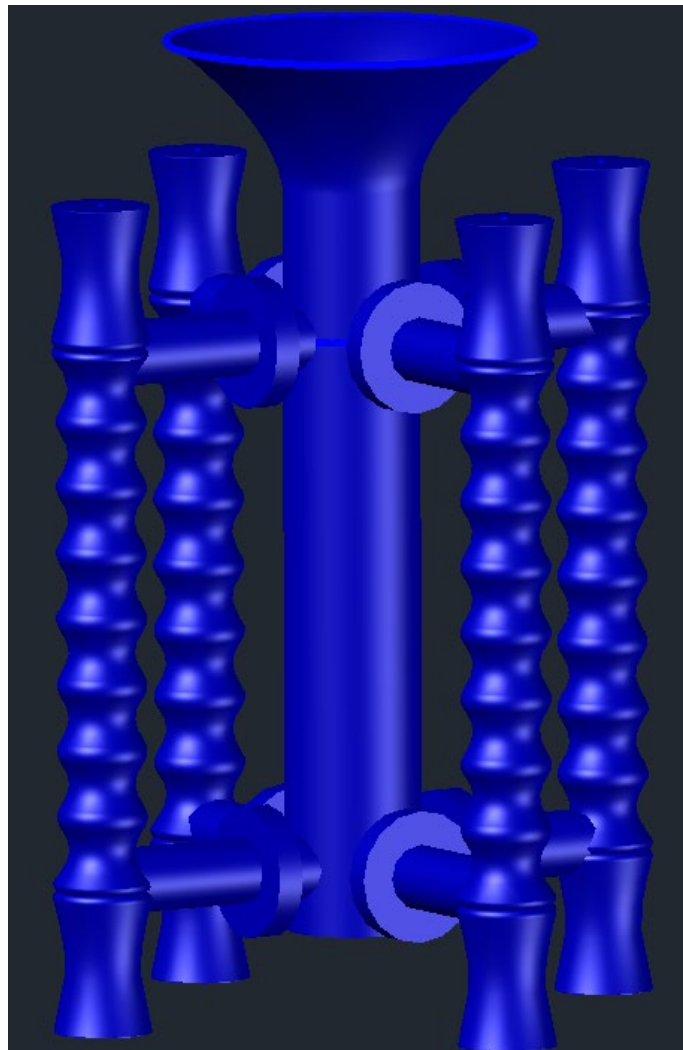


Figure 23: 3D CAD drawing of the wax tree

4 HEAT TREATMENT PLAN

From the heat treatment methods and cold working methods described in research & analysis section following method is selected to execute the heat treatment plan:

1. Solution Treatment (Annealing)

let's look how to perform heat treatment.

Suggested heat treatment plan for the door handle:

Solution Treatment : After casting, carry out solution treatment. This process results in a uniform steel structure and eliminates work hardening, making subsequent cold processing easier. It also restores the inherent corrosion resistance of the stainless steel, thereby enhancing its corrosion resistance. Additionally, this process can help increase formability.

The details of each step are explained further below.

Solution Treatment :

Effects:

1. Dissolution of Alloy Carbides in Stainless Steel & prevent reprecipitation.
 - Carbon (C) is one of the alloying elements present in steel. While it has a slight strengthening effect, it is detrimental to corrosion resistance, especially when it forms carbides with chromium (Cr).
 - To minimize the existence of C and Cr carbides, the solubility of C in austenite is manipulated by heating and cooling.
 - The solubility of C in austenite is high at high temperatures (0.34% at 1200°C) and low at low temperatures (0.02% at 600°C, and even lower at room temperature).
 - The steel is heated to a high temperature to dissolve the C-Cr compound and quickly cooled to prevent precipitation.
 - This helps to improve the steel's corrosion resistance, especially its intergranular corrosion resistance.
2. Prevent forming Sigma (σ) Phase (non-magnetic intermetallic phase composed mainly of iron and chromium)
 - Long-term heating in the range of 500-900°C can result in the precipitation of σ phase in austenitic steel.

- This increases the steel's brittleness and decreases its corrosion resistance. And also it causes to loss of ductility, toughness.
- The σ phase can be eliminated by dissolving it at a temperature higher than its precipitation temperature and rapidly cooling it to prevent re-precipitation.

Process:

- **Heating:** Begin by heating the door handle to a temperature within the range of 1000-1150°C, typically around 1020-1080°C. This temperature range is recommended by the GB1200 standard and is suitable for the specific composition of 304 stainless steel. The purpose of this heating step is to allow carbide and various alloying elements to dissolve in austenite fully and uniformly, creating a homogeneous structure.
- **Holding:** Maintain this temperature for enough time to ensure complete dissolution of carbides. The exact holding time can vary depending on the size and shape of the door handle, but it's crucial to allow enough time (60 minutes – according to GB1200 standard) for the heat to penetrate completely through the material.
- **Cooling:** After the holding period, cool the door handle quickly to prevent carbide precipitation. The method of cooling should be chosen based on the carbon content and size of the steel:
 - For steel with a carbon content greater than or equal to 0.08%, or if the chromium content is greater than 22% and there's a significant amount of nickel, **water cooling** should be used.
 - For steel with a carbon content less than 0.08% and a size greater than 3mm, **air cooling** is suitable.
 - For steel with a carbon content less than 0.08% and a size less than or equal to 0.5mm, **air cooling** can also be used.

Therefore, suitable cooling method for our sample is “Water Cooling”.

5 CONCLUSION

This research emphasizes the critical aspects of door handle design, highlighting the careful balance required between functionality, aesthetics, and practicality. Material selection is a crucial consideration, with stainless steel identified as an ideal choice due to its strength, durability, and versatility. Various types of stainless steel, including 304 stainless steels, are explored for their corrosion resistance, mechanical properties, and aesthetic appeal.

The design process incorporates ergonomic factors, cultural connotations, and stylistic influences to create a door handle that transcends mere functionality, aiming to be a piece of functional art enhancing the architectural space. The focus on stainless steel is strategic, considering factors such as formability, cost-effectiveness, and user comfort. The chosen 304 stainless steel, known for its corrosion resistance and heat resistance, aligns with hygiene requirements for door handles.

Metal casting methods, such as investment casting, are detailed, emphasizing precision, low material waste, and an excellent surface finish. The meticulous design of the stainless-steel door handle prioritizes comfort, aesthetics, and strength. Lever-style pull handles, polished finishes, and internal reinforcements contribute to a high-quality product. The manufacturing process involves 3D-printed wax patterns, ceramic shell creation, molten stainless-steel pouring, cutting and finishing and transfer to heat treatment processes.

The final step in optimizing the properties of a general-use door handle crafted from 304 stainless steel is the Solution Treatment, a critical process performed subsequent to casting. This specialized annealing procedure is designed to enhance the door handle's corrosion resistance by facilitating the dissolution of alloy carbides within the stainless-steel structure. By subjecting the material to carefully controlled heat, we promote the homogenization of the alloy, ensuring a more uniform distribution of elements. This not only fortifies the overall corrosion resistance of the door handle but also contributes to an improvement in hardness, toughness, and dimensional stability. The meticulous execution of Solution Treatment plays a pivotal role in achieving a door handle with superior material characteristics, setting the foundation for durability and reliability in everyday use.

6 REFERENCES

- [1] “The door handle as a critical interface with the structure and the material of the building,” in *Architectural Digest*, 2023.
- [2] J. Smith, “The importance of ergonomics in handle design,” in *Journal of Ergonomics*, vol. 45, no. 2, pp. 123-134, 2023.
- [3] M. Johnson, “The cultural and historical significance of door handles,” in *Cultural Anthropology*, vol. 33, no. 4, pp. 567-578, 2023.
- [4] L. Davis, “The stylistic influences in door handle design,” in *Interior Design Magazine*, 2023.
- [5] admin, “Door Handle Materials - Understanding The Pros and Cons.,” *Luxterior*, Sep. 16, 2021. <https://luxterior.com.au/door-handle-materials-understanding-the-pros-and-cons/>
- [6] M. Marketing, “Comparing Door Handle Materials: What’s Best for Your Needs?,” *The Lock Shop*, Sep. 25, 2023. <https://thelockshop.com.au/blogs/news/comparing-door-handle-material> (accessed Nov. 25, 2023).
- [7] A. Admin, “What metals are door handles made of? – Atlantic UK Architectural Hardware,” Jun. 20, 2023. <https://atlantichandles.com/what-metals-are-door-handles-made-of/> (accessed Nov. 25, 2023).
- [8] J. C. Casting, “4 Types Stainless Steel Heat Treatment Process | Cautions,” *www.investmentcastingpci.com*, Aug. 21, 2019. <https://www.investmentcastingpci.com/blog/stainless-steel-heat-treatment-process-and-cautions/>
- [9] “Stainless Steel - Heat Treatment,” *AZoM.com*, Jan. 02, 2002. <https://www.azom.com/article.aspx?ArticleID=1141>
- [10] S. Tukur, A. Ahmed, and N. Mu'az, “Effect of Heat Treatment Temperature on Mechanical Properties of the AISI 304 Stainless Steel,” *International Journal of Innovative Research in Science, Engineering and Technology (An ISO)*, vol. 3297, no. 2, pp. 2319–8753, 2007, Available: <https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=e756ae703518629652a41efd7a22f3f0f259df39>
- [11] Mohammad Tavakoli Bina, F. Dehghani, and M. Salimi, “Effect of heat treatment on bonding interface in explosive welded copper/stainless steel,” vol. 45, pp. 504–509, Mar. 2013, doi: <https://doi.org/10.1016/j.matdes.2012.09.037>.
- [12] V. Shamasundar, Gopalakrishna, and Janagan, “Gravity Die Casting Process Die Design and Process Optimisation.” Available: https://www.esi-group.com/sites/default/files/resource/publication/1294/f_gdc_ennore.pdf
- [13] “Gravity Die Casting - What Is Gravity Die Casting, Gravity Die Cast Process,” *Sunrise Metal - Aluminium Die Casting Expert*. <https://www.sunrise-metal.com/what-is-gravity-die-casting/>
- [14] “Pressure Die Casting,” *www.themetalcasting.com*. <https://www.themetalcasting.com/pressure-die-casting.html>
- [15] “What Is the High Pressure Die Casting (HPDC) Process | RapidDirect Blog,” *Rapid Direct*, May 31, 2022. <https://www.rapiddirect.com/blog/what-is-high-pressure-die-casting/>

[16]Rapiddirect, “Centrifugal Casting: Types, Processes, and Industrial Applications,” *rapiddirect*, Jul. 27, 2022. <https://www.rapiddirect.com/blog/centrifugal-casting/>

[17]“What is Centrifugal Casting and How Does It Work?,” *MetalTek*.
<https://www.metaltek.com/blog/what-is-centrifugal-casting/>

[18]S. Marolt, “Benefits of stainless steel door handles for entrance doors - Griffing: Inox ročaji in kljuke za vhodna vrata,” Griffing: Inox Ročaji in Kljuke Za Vhodna Vrata -, Apr. 20, 2021.
<https://www.griffing.si/blog-post/benefits-of-stainless-steel-door-handles-for-external-doors/?lang=en>

[19]“The benefits of stainless steel door handles,” StainlessDoorHardware.com.
<https://www.stainlessdoorhardware.com/blog/the-benefits-of-stainless-steel-door-handles/>

[20]“Care & maintenance of stainless steel door handles.” <https://www.alpro.co.uk/care-and-maintenance>

[21]Cmostores.com limited, “Why choose grade 304 stainless steel door handles? - Door Superstore Help & Advice,” Door Superstore Help & Advice, Feb. 24, 2023.
<https://www.doorsuperstore.co.uk/help-and-advice/product-guides/door-furniture/why-choose-grade-304-stainless-steel-door-handles/#:~:text=304%20grade%20stainless%20steel%20has,corrosion%20resistance%2C%20durability%20and%20hardness.>

[22]I. Doors, “4 Major benefits of stainless steel door handles - Iron Doors Plus, Inc.,” Iron Doors Plus, Inc., Mar. 15, 2022. <https://irondoorsplus.com/4-major-benefits-of-stainless-steel-door-handles/>

[23]“Investment Casting: Know Its Process, Benefits, Applications,” *rapiddirect*, Sep. 11, 2022.
<https://www.rapiddirect.com/blog/what-is-investment-casting/>

[24] The Ultimate Guide to Stainless Steel Heat Treatment <https://www.machinemfg.com/stainless-steel-heat-treatment/>

[25]Stress relieving heat treatments for austenitic stainless steels – BSSA
https://bssa.org.uk/bssa_articles/stress-relieving-heat-treatments-for-austenitic-stainless-steels/

[26]Heat Treatments for Austenitic Stainless Steels <https://www.gpss.com/heat-treatments/>

[27]Stainless Steel Heat Treatment Options - Engineer Excel <https://engineerexcel.com/stainless-steel-heat-treatment/>

[28]<https://www.facebook.com/machmfg>, “The Ultimate Guide to Stainless Steel Heat Treatment | MachineMfg,” *www.machinemfg.com*, Jun. 11, 2023. <https://www.machinemfg.com/stainless-steel-heat-treatment/>